

An empirical example: Four terminals and two areas

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```
# Four taxa and two areas
```

```
## Libraries
```

```
library(blepd)
```

```
## Loading required package: ape
```

```
## Loading required package: picante
```

```
## Loading required package: vegan
```

```
## Loading required package: permute
```

```
## Loading required package: lattice
```

```
## This is vegan 2.6-6.1
```

```
## Loading required package: nlme
```

```
library(ggtree)
```

```
## ggtree v3.10.1 For help: https://yulab-smu.top/treedata-book/
```

```
##
```

```
## If you use the ggtree package suite in published research, please cite  
## the appropriate paper(s):
```

```
##
```

```
## Guangchuan Yu, David Smith, Huachen Zhu, Yi Guan, Tommy Tsan-Yuk Lam.
```

```
## ggtree: an R package for visualization and annotation of phylogenetic
```

```
## trees with their covariates and other associated data. Methods in
```

```
## Ecology and Evolution. 2017, 8(1):28-36. doi:10.1111/2041-210X.12628
```

```
##
```

```
## S Xu, Z Dai, P Guo, X Fu, S Liu, L Zhou, W Tang, T Feng, M Chen, L
```

```
## Zhan, T Wu, E Hu, Y Jiang, X Bo, G Yu. ggtreeExtra: Compact
```

```
## visualization of richly annotated phylogenetic data. Molecular Biology
```

```
## and Evolution. 2021, 38(9):4039-4042. doi: 10.1093/molbev/msab166
```

```
##
```

```
## LG Wang, TTY Lam, S Xu, Z Dai, L Zhou, T Feng, P Guo, CW Dunn, BR
```

```
## Jones, T Bradley, H Zhu, Y Guan, Y Jiang, G Yu. treeio: an R package
```

```
## for phylogenetic tree input and output with richly annotated and
```

```
## associated data. Molecular Biology and Evolution. 2020, 37(2):599-603.
```

```
## doi: 10.1093/molbev/msz240
```

```
##
```

```
## Attaching package: 'ggtree'
```

```

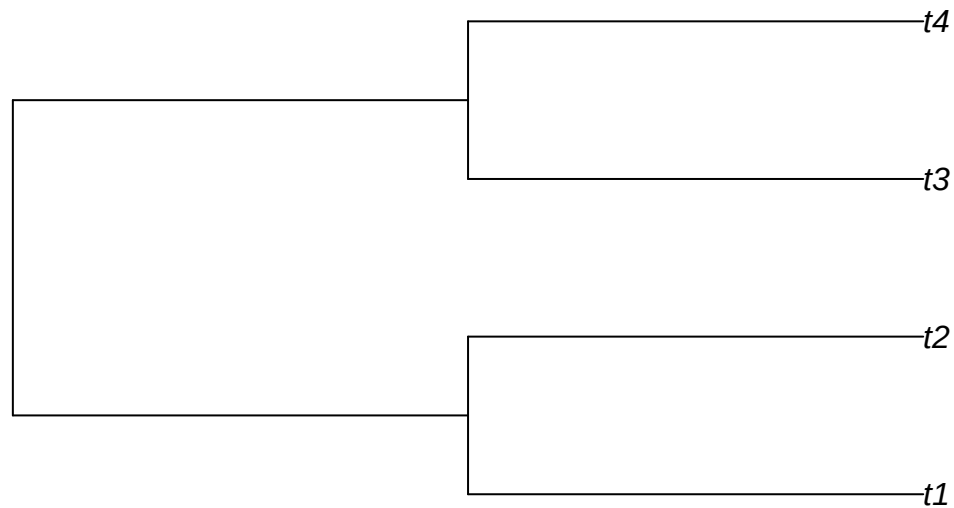
## The following object is masked from 'package:nlme':
##
##      collapse
## The following object is masked from 'package:ape':
##
##      rotate
library(ggplot2)

## Data
# Call the 4 terminals example
data(package = "blepd")

## trees
data(tree)
str(tree)

## List of 5
## $ edge      : int [1:6, 1:2] 5 6 6 5 7 7 6 1 2 7 ...
## $ edge.length: num [1:6] 1 1 1 1 1 1
## $ Nnode      : int 3
## $ tip.label  : chr [1:4] "t1" "t2" "t3" "t4"
## $ root.edge  : num 1
## - attr(*, "class")= chr "phylo"
## - attr(*, "order")= chr "cladewise"
plot(tree,use.edge.length =T)

```



```

initialTree <- tree

## distributions
data(distribution)
str(distribution)

## int [1:2, 1:4] 1 0 0 1 1 0 0 1
## - attr(*, "dimnames")=List of 2
## ..$ : chr [1:2] "A1" "A2"
## ..$ : chr [1:4] "t1" "t2" "t3" "t4"
dist4taxa <- distribution

## distribution to XY to be able to plot the data, otherwise, skip the next step
distXY <- matrix2XY(dist4taxa)

distXY

## Terminal Area
## 1      t1    A1
## 2      t3    A1
## 3      t2    A2
## 4      t4    A2

```

```
## plotting

## the tree

plotTree <- ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=5, color="black") +
  theme_tree2() +
  ggtitle("Four terminals, all branches are == 1")

##
print(plotTree)
```

Four terminals, all branches are == 1



```
## the distribution

plotDistrib <- ggplot(data=distXY,
                      aes(x= Area, y= Terminal , size =150)) +
  geom_point() +
  theme_bw() +
  theme( axis.text.y = element_blank(),
          panel.grid.major = element_blank(), # Remove major gridlines
          panel.grid.minor = element_blank(), # Remove minor gridlines
          axis.line = element_line(colour = NA_character_), # Remove axis lines
          axis.ticks = element_blank(), # Remove axis ticks
```

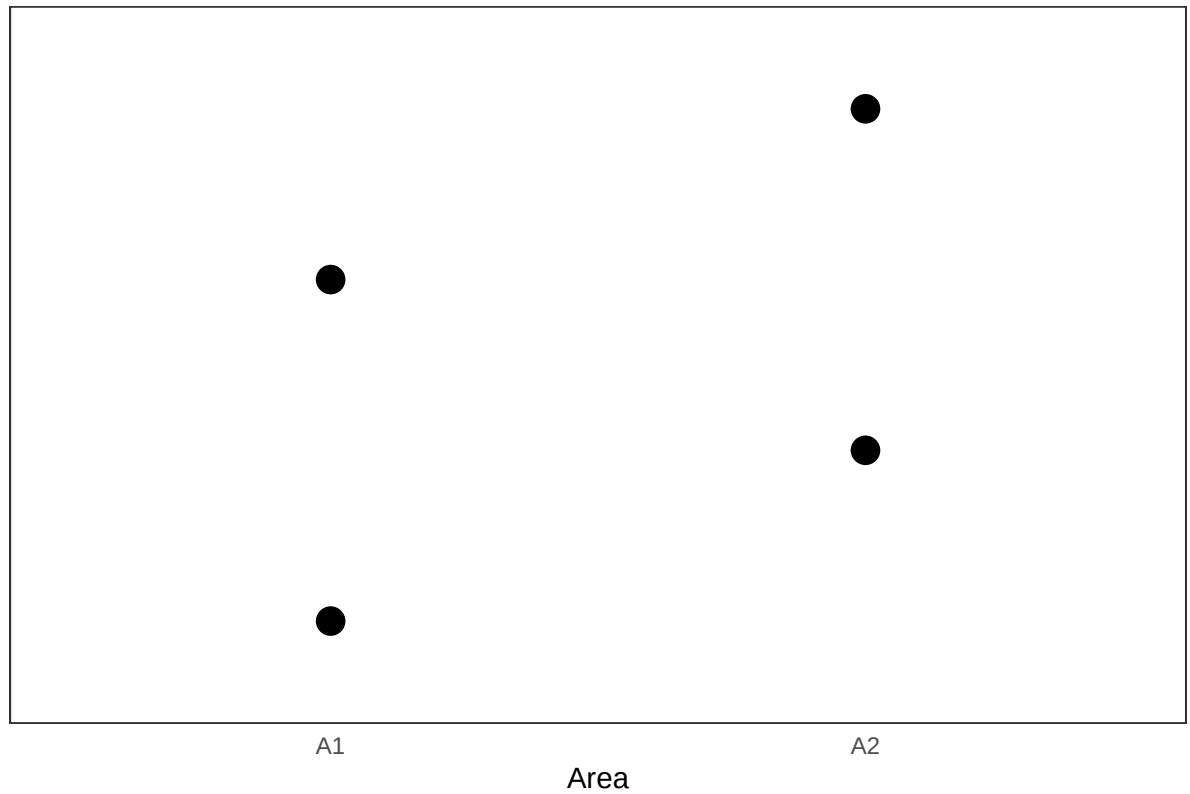
```

        legend.position = "none"
      ) +
    labs(title = "Distributions",
         y = "",
         x = "Area")

##
print(plotDistrib)

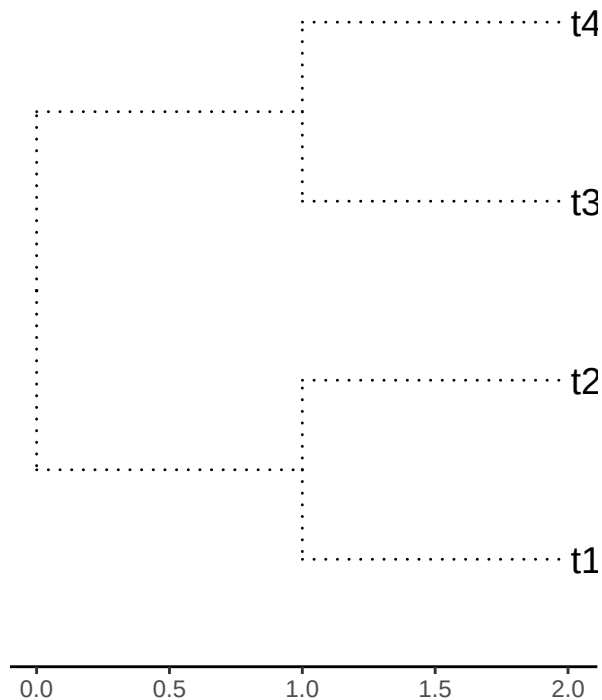
```

Distributions

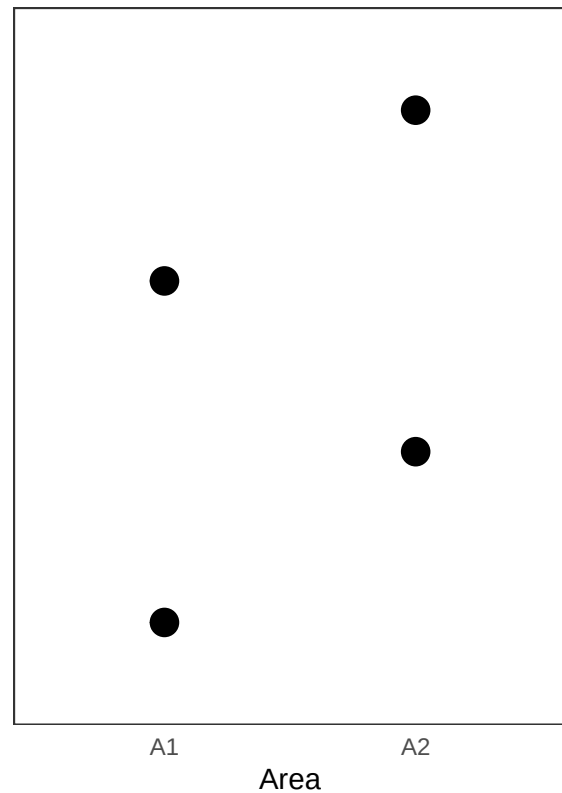


```
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)
```

Four terminals, all branches are == 1



Distributions



```
#We check whether names in both objects: initialTree and dist4taxa, are the same.
```

```
##all(colnames(dist4taxa) == initialTree$tip.label)
```

```
all(colnames(dist4taxa) %in% initialTree$tip.label)
```

```
## [1] TRUE
```

```
#We report the branch length, and calculate the PD values.
```

```
initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)
```

```
initialPD
```

```
## [1] 4 4
```

```
#As expected there is a tie between both areas.
```

```
#Now, we can see the impact of swapping the branch lengths; we will use the three  
#available models to swap: "simpleswap", "allswap", "uniform", and we will swap the  
#three classes of branches: "terminals", "internals", "all".
```

?swapBL

```
for( modelo in c("simpleswap","allswap","uniform") ){  
  for( rama in c("terminals","internals","all") ){  
    val <- swapBL(tree=initialTree,  
                  distribution = dist4taxa,  
                  model = modelo,  
                  branch = rama  
                  )  
  
    cat( "\n\t Model=\t\t",modelo,  
        "\n\t Branchs swapped=\t",rama,"\n" )  
  
    printBlepd(val)  
  
    cat( "\n\n\n" )  
  }  
}
```

```
## model to test simpleswap reps 100  
##  
##   Model=      simpleswap  
##   Branchs swapped=  terminals  
##  
## BestInitial:A1A2  
##   AreaSelected Freq Percent  
## 1           A1A2 100      100  
##  
##  
##  
## model to test simpleswap reps 100  
##  
##   Model=      simpleswap  
##   Branchs swapped=  internals  
##  
## BestInitial:A1A2  
##   AreaSelected Freq Percent  
## 1           A1A2 100      100  
##  
##  
##  
## model to test simpleswap reps 100  
##  
##   Model=      simpleswap  
##   Branchs swapped=  all  
##  
## BestInitial:A1A2  
##   AreaSelected Freq Percent
```

```

## 1          A1A2  100    100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1          A1A2  100    100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1          A1A2  100    100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1          A1A2  100    100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1          A1A2  100    100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  internals
##
## BestInitial:A1A2

```



```
## AreaSelected Freq Percent
## 1          A1A2 100      100
```

```
##
##
##
```

```
## model to test uniform reps 100
```

```
##
```

```
## Model=      uniform
```

```
## Branchs swapped=    all
```

```
##
```

```
## BestInitial:A1A2
```

```
## AreaSelected Freq Percent
```

```
## 1          A1A2 100      100
```

```
##As expected, as all braches are EQUAL nothing happens
```

```
##Now, let us see if the results could be assigned to the longest branches (or not).
```

```
## we could use a single command (*evalBranch*)
```

```
?evalBranch
```

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")
```

```
## Loading required package: maps
```

```
##
```

```
## Attaching package: 'phytools'
```

```
## The following object is masked from 'package:vegan':
```

```
##
```

```
##      scores
```

```
## let's see the structure of the output
```

```
str(AllTerminals)
```

```
## List of 2
```

```
## $ lower:List of 4
```

```
## ..$ : 'data.frame': 1 obs. of 5 variables:
```

```
## .. ..$ branchToEval: chr "[t1]|1"
```

```
## .. ..$ resp : chr "A1A2"
```

```
## .. ..$ resp.1 : chr "A2"
```

```
## .. ..$ approach : chr "lower"
```

```
## .. ..$ delta : num -1.02
```

```
## ..$ : 'data.frame': 1 obs. of 5 variables:
```

```
## .. ..$ branchToEval: chr "[t2]|2"
```

```
## .. ..$ resp : chr "A1A2"
```

```
## .. ..$ resp.1 : chr "A1"
```

```
## .. ..$ approach : chr "lower"
```

```
## .. ..$ delta      : num -1.02
## ..$ : 'data.frame': 1 obs. of  5 variables:
## .. ..$ branchToEval: chr "[t3]|3"
## .. ..$ resp        : chr "A1A2"
## .. ..$ resp.1      : chr "A2"
## .. ..$ approach    : chr "lower"
## .. ..$ delta       : num -1.02
## ..$ : 'data.frame': 1 obs. of  5 variables:
## .. ..$ branchToEval: chr "[t4]|4"
## .. ..$ resp        : chr "A1A2"
## .. ..$ resp.1      : chr "A1"
## .. ..$ approach    : chr "lower"
## .. ..$ delta       : num -1.02
## $ upper:List of 4
## ..$ : 'data.frame': 1 obs. of  5 variables:
## .. ..$ branchToEval: chr "[t1]|1"
## .. ..$ resp        : chr "A1A2"
## .. ..$ resp.1      : chr "A1"
## .. ..$ approach    : chr "upper"
## .. ..$ delta       : num 2.07
## ..$ : 'data.frame': 1 obs. of  5 variables:
## .. ..$ branchToEval: chr "[t2]|2"
## .. ..$ resp        : chr "A1A2"
## .. ..$ resp.1      : chr "A2"
## .. ..$ approach    : chr "upper"
## .. ..$ delta       : num 2.07
## ..$ : 'data.frame': 1 obs. of  5 variables:
## .. ..$ branchToEval: chr "[t3]|3"
## .. ..$ resp        : chr "A1A2"
## .. ..$ resp.1      : chr "A1"
## .. ..$ approach    : chr "upper"
## .. ..$ delta       : num 2.07
## ..$ : 'data.frame': 1 obs. of  5 variables:
## .. ..$ branchToEval: chr "[t4]|4"
## .. ..$ resp        : chr "A1A2"
## .. ..$ resp.1      : chr "A2"
## .. ..$ approach    : chr "upper"
## .. ..$ delta       : num 2.07
```

Now we can print a far more friendly output

```
printEvalBranch(AllTerminals, compact=FALSE)
```

```
##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1      A1A2      A2   lower -1.024
## 2 [t2]|2      A1A2      A1   lower -1.024
## 3 [t3]|3      A1A2      A2   lower -1.024
## 4 [t4]|4      A1A2      A1   lower -1.024
## 5 [t1]|1      A1A2      A1   upper  2.072
## 6 [t2]|2      A1A2      A2   upper  2.072
## 7 [t3]|3      A1A2      A1   upper  2.072
## 8 [t4]|4      A1A2      A2   upper  2.072
```

```
## Now the internals
```

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")
```

```
printEvalBranch(AllTerminals, compact=FALSE)
```

```
##           node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2] |6      A1A2      A1A2  lower      0
## 2 [node number:7:t3/t4] |7      A1A2      A1A2  lower      0
## 3 [node number:6:t1/t2] |6      A1A2      A1A2  upper      0
## 4 [node number:7:t3/t4] |7      A1A2      A1A2  upper      0
```

```
## Now all branches
```

```
Allall <- evalBranch(tree=initialTree,
                    distribution = dist4taxa,
                    branchToEval="all",
                    approach="all")
```

```
printEvalBranch(Allall, compact=FALSE)
```

```
##           node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2] |6      A1A2      A1A2  lower      0
## 2 [t1] |1      A1A2      A2    lower -1.024
## 3 [t2] |2      A1A2      A1    lower -1.024
## 4 [node number:7:t3/t4] |7      A1A2      A1A2  lower      0
## 5 [t3] |3      A1A2      A2    lower -1.024
## 6 [t4] |4      A1A2      A1    lower -1.024
## 7 [node number:6:t1/t2] |6      A1A2      A1A2  upper      0
## 8 [t1] |1      A1A2      A1    upper  2.072
## 9 [t2] |2      A1A2      A2    upper  2.072
## 10 [node number:7:t3/t4] |7      A1A2      A1A2  upper      0
## 11 [t3] |3      A1A2      A1    upper  2.072
## 12 [t4] |4      A1A2      A2    upper  2.072
```

```
#now with a longer branch
```

```
initialTree <- tree
```

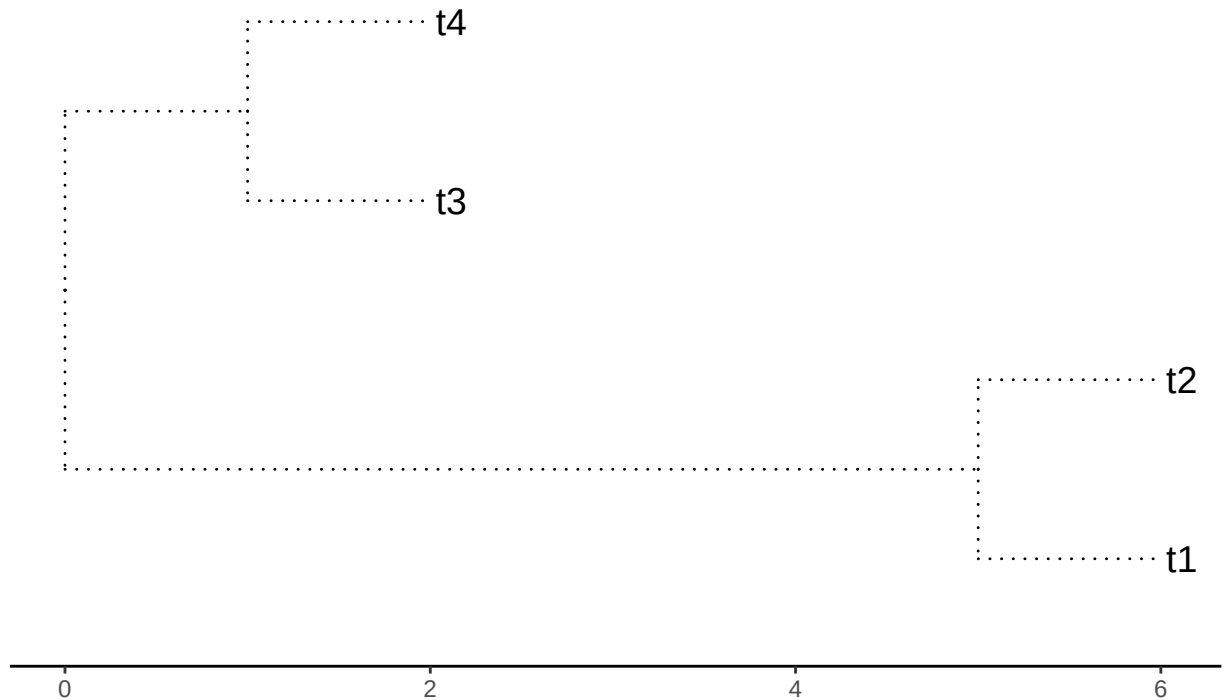
```
initialTree$edge.length[1] <- 5
```

```
plotTree <- ggtree(initialTree, ladderize=TRUE,
                  color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=5, color="black") +
  theme_tree2() +
  ggtitle("Four terminals branch 1 == 5")
```

```
##
```

```
print(plotTree)
```

Four terminals branch 1 == 5



```
for( modelo in c("simpleswap","allswap","uniform") ){  
  for( rama in c("terminals","internals","all") ){  
    val <- swapBL(tree=initialTree,  
                  distribution = dist4taxa,  
                  model = modelo,  
                  branch = rama  
                  )  
  
    cat( "\n\t Model=\t\t",modelo,  
        "\n\t Branchs swapped=\t",rama,"\n" )  
  
    printBlepd(val)  
  
    cat( "\n\n\n" )  
  }  
}
```

```
## model to test simpleswap reps 100  
##  
## Model=      simpleswap
```

```

## Branchs swapped= terminals
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1          A1A2 100      100
##
##
##
## model to test simpleswap reps 100
##
## Model=      simpleswap
## Branchs swapped= internals
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1          A1A2 100      100
##
##
##
## model to test simpleswap reps 100
##
## Model=      simpleswap
## Branchs swapped= all
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1          A1   17      17
## 2          A1A2  70      70
## 3          A2   13      13
##
##
##
## model to test allswap reps 100
##
## Model=      allswap
## Branchs swapped= terminals
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1          A1A2 100      100
##
##
##
## model to test allswap reps 100
##
## Model=      allswap
## Branchs swapped= internals
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1          A1A2 100      100
##
##
##

```

```

## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=    all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1    29      29
## 2          A1A2    36      36
## 3           A2    35      35
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1          A1A2   100     100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1          A1A2   100     100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=    all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1    50      50
## 2           A2    50      50

```

```

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

```

```

##   node initialArea FinalArea Aproach %Delta
## 1 [t1]|1          A1A2          A2   lower -1.024
## 2 [t2]|2          A1A2          A1   lower -1.024

```

```
## 3 [t3]|3      A1A2      A2   lower -1.024
## 4 [t4]|4      A1A2      A1   lower -1.024
## 5 [t1]|1      A1A2      A1   upper  2.066
## 6 [t2]|2      A1A2      A2   upper  2.066
## 7 [t3]|3      A1A2      A1   upper  2.066
## 8 [t4]|4      A1A2      A2   upper  2.066
```

Now the internals

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")
```

```
printEvalBranch(AllTerminals, compact=FALSE)
```

```
##           node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6      A1A2      A1A2   lower      0
## 2 [node number:7:t3/t4]|7      A1A2      A1A2   lower      0
## 3 [node number:6:t1/t2]|6      A1A2      A1A2   upper      0
## 4 [node number:7:t3/t4]|7      A1A2      A1A2   upper      0
```

Now all branches

```
Allall <- evalBranch(tree=initialTree,
                    distribution = dist4taxa,
                    branchToEval="all",
                    approach="all")
```

```
printEvalBranch(Allall, compact=FALSE)
```

```
##           node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6      A1A2      A1A2   lower      0
## 2 [t1]|1      A1A2      A2   lower -1.024
## 3 [t2]|2      A1A2      A1   lower -1.024
## 4 [node number:7:t3/t4]|7      A1A2      A1A2   lower      0
## 5 [t3]|3      A1A2      A2   lower -1.024
## 6 [t4]|4      A1A2      A1   lower -1.024
## 7 [node number:6:t1/t2]|6      A1A2      A1A2   upper      0
## 8 [t1]|1      A1A2      A1   upper  2.066
## 9 [t2]|2      A1A2      A2   upper  2.066
## 10 [node number:7:t3/t4]|7      A1A2      A1A2   upper      0
## 11 [t3]|3      A1A2      A1   upper  2.066
## 12 [t4]|4      A1A2      A2   upper  2.066
```

#now with a longer branch

```
initialTree <- tree
```

```
initialTree$edge.length[1] <- 10
```

```
plotTree <- ggtree(initialTree, ladderize=TRUE,
```

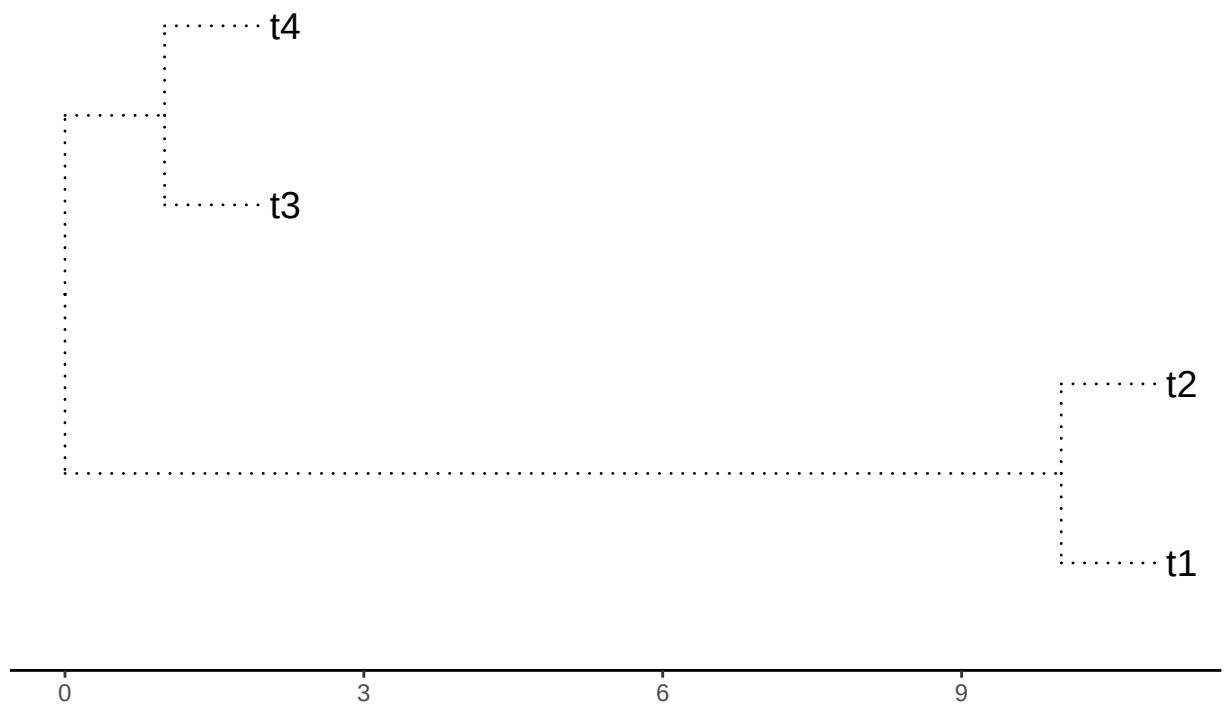
```

        color="black", size=0.5, linetype="dotted") +
geom_tiplab(size=5, color="black") +
theme_tree2() +
ggtitle("Four terminals branch 1 == 10")

##
print(plotTree)

```

Four terminals branch 1 == 10



```

for( modelo in c("simpleswap","allswap","uniform") ){

  for( rama in c("terminals","internals","all") ){

    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                  )

    cat( "\n\t Model=\t\t",modelo,
          "\n\t Branchs swapped=\t",rama,"\n" )

    printBlepd(val)

    cat( "\n\n\n" )
  }
}

```



```

    }
}

```

```

## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1   12      12
## 2           A1A2 79      79
## 3           A2    9       9
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  internals

```

```

##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=    all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1    39      39
## 2           A1A2   34      34
## 3           A2    27      27
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=    all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1    51      51
## 2           A2    49      49

```

```

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,

```

```

        branchToEval="terminals",
        approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1      A1A2      A2   lower -1.024
## 2 [t2]|2      A1A2      A1   lower -1.024
## 3 [t3]|3      A1A2      A2   lower -1.024
## 4 [t4]|4      A1A2      A1   lower -1.024
## 5 [t1]|1      A1A2      A1   upper  2.032
## 6 [t2]|2      A1A2      A2   upper  2.032
## 7 [t3]|3      A1A2      A1   upper  2.032
## 8 [t4]|4      A1A2      A2   upper  2.032

## Now the internals

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

##      node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6      A1A2      A1A2   lower    0
## 2 [node number:7:t3/t4]|7      A1A2      A1A2   lower    0
## 3 [node number:6:t1/t2]|6      A1A2      A1A2   upper    0
## 4 [node number:7:t3/t4]|7      A1A2      A1A2   upper    0

## Now all branches

Allall <- evalBranch(tree=initialTree,
                     distribution = dist4taxa,
                     branchToEval="all",
                     approach="all")

printEvalBranch(Allall, compact=FALSE)

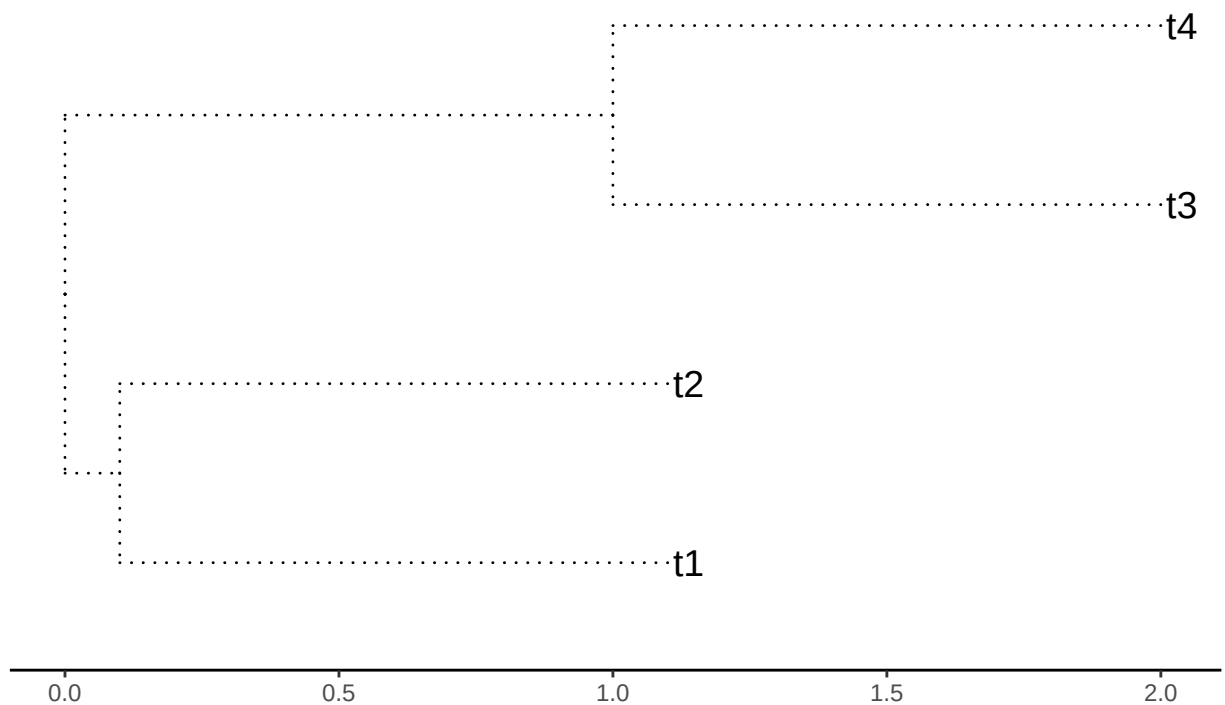
##      node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6      A1A2      A1A2   lower    0
## 2      [t1]|1      A1A2      A2   lower -1.024
## 3      [t2]|2      A1A2      A1   lower -1.024
## 4 [node number:7:t3/t4]|7      A1A2      A1A2   lower    0
## 5      [t3]|3      A1A2      A2   lower -1.024
## 6      [t4]|4      A1A2      A1   lower -1.024
## 7 [node number:6:t1/t2]|6      A1A2      A1A2   upper    0
## 8      [t1]|1      A1A2      A1   upper  2.032
## 9      [t2]|2      A1A2      A2   upper  2.032
## 10 [node number:7:t3/t4]|7      A1A2      A1A2   upper    0
## 11      [t3]|3      A1A2      A1   upper  2.032
## 12      [t4]|4      A1A2      A2   upper  2.032

```

#now with a longer branch

```
initialTree <- tree  
initialTree$edge.length[1] <- 0.1  
  
plotTree <- ggtree(initialTree, ladderize=TRUE,  
                    color="black", size=0.5, linetype="dotted") +  
  geom_tiplab(size=5, color="black") +  
  theme_tree2() +  
  ggtitle("Four terminals branch 1 == 0.1")  
  
##  
print(plotTree)
```

Four terminals branch 1 == 0.1



```
for( modelo in c("simpleswap", "allswap", "uniform") ){  
  for( rama in c("terminals", "internals", "all") ){  
    val <- swapBL(tree=initialTree,  
                  distribution = dist4taxa,  
                  model = modelo,
```

```

        branch = rama
    )

    cat( "\n\t Model=\t\t",modelo,
        "\n\t Branchs swapped=\t",rama,"\n" )

    printBlepd(val)

    cat( "\n\n\n" )

}

}

```

```

## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2 100      100
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1   17       17
## 2           A1A2 69       69
## 3           A2   14       14
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  terminals
##

```

```

## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2  100     100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2  100     100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  all
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1    32     32
## 2           A1A2  29     29
## 3           A2    39     39
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  terminals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2  100     100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  internals
##
## BestInitial:A1A2
##   AreaSelected Freq Percent
## 1           A1A2  100     100
##
##
##
## model to test uniform reps 100
##

```

```
## Model= uniform
## Branchs swapped= all
##
## BestInitial:A1A2
## AreaSelected Freq Percent
## 1 A1 50 50
## 2 A2 50 50

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
## node initialArea FinalArea Approach %Delta
## 1 [t1]|1 A1A2 A2 lower -1.024
## 2 [t2]|2 A1A2 A1 lower -1.024
## 3 [t3]|3 A1A2 A2 lower -1.024
## 4 [t4]|4 A1A2 A1 lower -1.024
## 5 [t1]|1 A1A2 A1 upper 2.061
## 6 [t2]|2 A1A2 A2 upper 2.061
## 7 [t3]|3 A1A2 A1 upper 2.061
## 8 [t4]|4 A1A2 A2 upper 2.061
```

Now the internals

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
## node initialArea FinalArea Approach %Delta
## 1 [node number:6:t1/t2]|6 A1A2 A1A2 lower 0
## 2 [node number:7:t3/t4]|7 A1A2 A1A2 lower 0
## 3 [node number:6:t1/t2]|6 A1A2 A1A2 upper 0
## 4 [node number:7:t3/t4]|7 A1A2 A1A2 upper 0
```

Now all branches

```
Allall <- evalBranch(tree=initialTree,
                    distribution = dist4taxa,
                    branchToEval="all",
                    approach="all")

printEvalBranch(Allall, compact=FALSE)
```

```
## node initialArea FinalArea Approach %Delta
## 1 [node number:6:t1/t2]|6 A1A2 A1A2 lower 0
## 2 [t1]|1 A1A2 A2 lower -1.024
## 3 [t2]|2 A1A2 A1 lower -1.024
## 4 [node number:7:t3/t4]|7 A1A2 A1A2 lower 0
```

```
## 5          [t3] | 3      A1A2      A2    lower -1.024
## 6          [t4] | 4      A1A2      A1    lower -1.024
## 7 [node number:6:t1/t2] | 6    A1A2    A1A2  upper      0
## 8          [t1] | 1      A1A2      A1    upper  2.061
## 9          [t2] | 2      A1A2      A2    upper  2.061
## 10 [node number:7:t3/t4] | 7    A1A2    A1A2  upper      0
## 11          [t3] | 3      A1A2      A1    upper  2.061
## 12          [t4] | 4      A1A2      A2    upper  2.061
```

#now with a longer branch in brach 2

#now with a longer branch

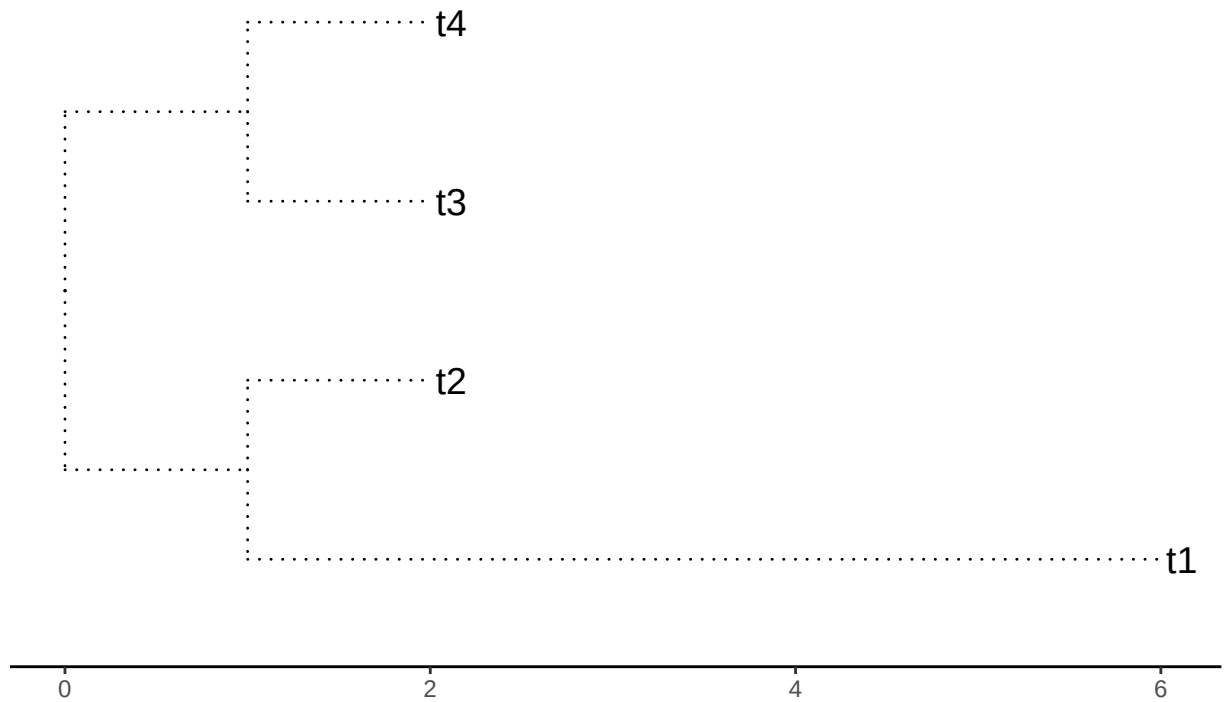
```
initialTree <- tree

initialTree$edge.length[2] <- 5

plotTree <- ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=5, color="black") +
  theme_tree2() +
  ggtitle("Four terminals branch 1 == 5")

##
print(plotTree)
```


Four terminals branch 1 == 5



```
for( modelo in c("simpleswap","allswap","uniform") ){
  for( rama in c("terminals","internals","all") ){
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )

    cat( "\n\t Model=\t\t",modelo,
          "\n\t Branchs swapped=\t",rama,"\n" )

    printBlepd(val)

    cat( "\n\n\n" )
  }
}
```

```
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  terminals
##
```

```

## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   67      67
## 2           A2   33      33
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=    internals
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1  100     100
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=    all
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   74      74
## 2          A1A2   12      12
## 3           A2   14      14
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=    terminals
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   51      51
## 2           A2   49      49
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=    internals
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1  100     100
##
##
##

```

```

## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=    all
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   36      36
## 2           A1A2  34      34
## 3           A2   30      30
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=    terminals
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   49      49
## 2           A2   51      51
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=    internals
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1  100     100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=    all
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   57      57
## 2           A2   43      43

```

```

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)

```

```

##   node initialArea FinalArea Aproach %Delta
## 1 [t1]|1           A1           A2   lower -80.193

```

```
## 2 [t2]|2      A1      A1  lower      0
## 3 [t3]|3      A1      A1  lower      0
## 4 [t4]|4      A1      A1  lower      0
## 5 [t1]|1      A1      A1  upper      0
## 6 [t2]|2      A1      A2  upper 405.005
## 7 [t3]|3      A1      A1  upper      0
## 8 [t4]|4      A1      A2  upper 405.005
```

Now the internals

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")
```

```
printEvalBranch(AllTerminals, compact=FALSE)
```

```
##              node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6      A1      A1  lower      0
## 2 [node number:7:t3/t4]|7      A1      A1  lower      0
## 3 [node number:6:t1/t2]|6      A1      A1  upper      0
## 4 [node number:7:t3/t4]|7      A1      A1  upper      0
```

Now all branches

```
Allall <- evalBranch(tree=initialTree,
                    distribution = dist4taxa,
                    branchToEval="all",
                    approach="all")
```

```
printEvalBranch(Allall, compact=FALSE)
```

```
##              node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6      A1      A1  lower      0
## 2 [t1]|1      A1      A2  lower -80.193
## 3 [t2]|2      A1      A1  lower      0
## 4 [node number:7:t3/t4]|7      A1      A1  lower      0
## 5 [t3]|3      A1      A1  lower      0
## 6 [t4]|4      A1      A1  lower      0
## 7 [node number:6:t1/t2]|6      A1      A1  upper      0
## 8 [t1]|1      A1      A1  upper      0
## 9 [t2]|2      A1      A2  upper 405.005
## 10 [node number:7:t3/t4]|7      A1      A1  upper      0
## 11 [t3]|3      A1      A1  upper      0
## 12 [t4]|4      A1      A2  upper 405.005
```

#now with a longer branch

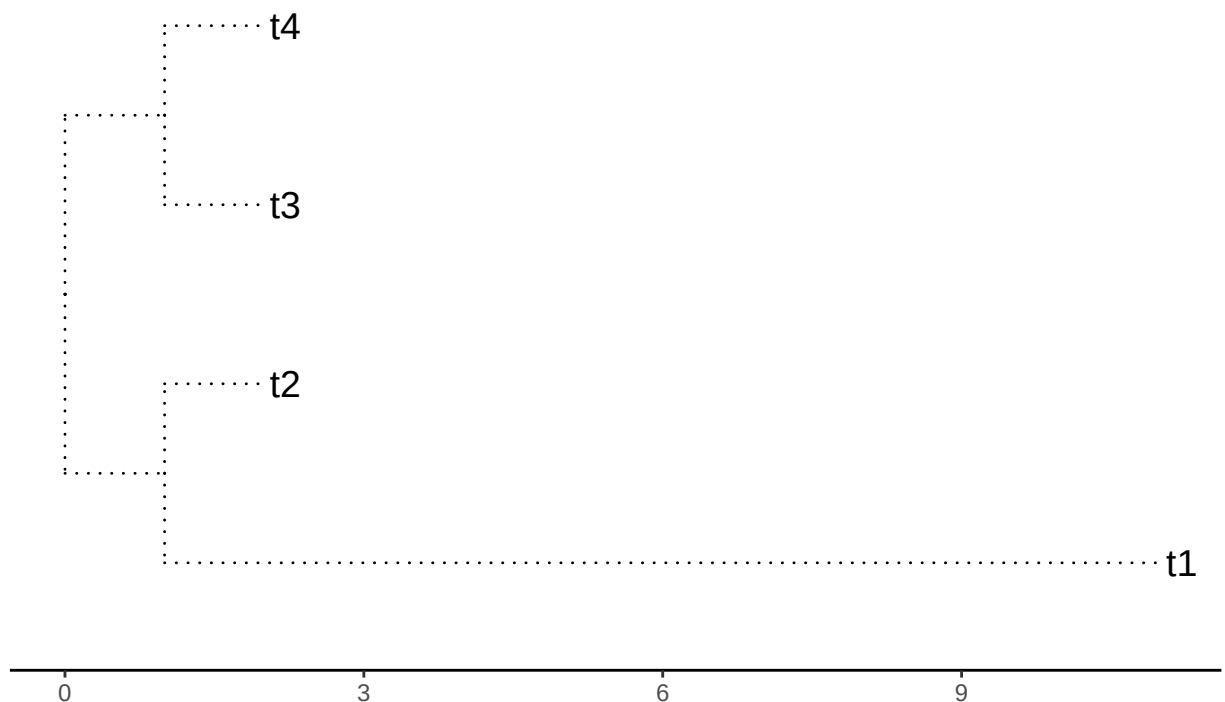
```
initialTree <- tree
```

```
initialTree$edge.length[2] <- 10
```

```
plotTree <- ggtree(initialTree, ladderize=TRUE,
                  color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=5, color="black") +
  theme_tree2() +
  ggtitle("Four terminals branch 1 == 10")

##
print(plotTree)
```

Four terminals branch 1 == 10



```
for( modelo in c("simpleswap", "allswap", "uniform") ){
  for( rama in c("terminals", "internals", "all") ){
    val <- swapBL(tree=initialTree,
                  distribution = dist4taxa,
                  model = modelo,
                  branch = rama
                )

    cat( "\n\t Model=\t\t", modelo,
          "\n\t Branchs swapped=\t", rama, "\n" )

    printBlepd(val)

    cat( "\n\n\n" )
  }
}
```

```
}  
  
}
```

```
## model to test simpleswap reps 100
```

```
##
```

```
## Model=      simpleswap
```

```
## Branchs swapped=  terminals
```

```
##
```

```
## BestInitial:A1
```

```
## AreaSelected Freq Percent
```

```
## 1          A1  73      73
```

```
## 2          A2  27      27
```

```
##
```

```
##
```

```
##
```

```
## model to test simpleswap reps 100
```

```
##
```

```
## Model=      simpleswap
```

```
## Branchs swapped=  internals
```

```
##
```

```
## BestInitial:A1
```

```
## AreaSelected Freq Percent
```

```
## 1          A1 100     100
```

```
##
```

```
##
```

```
##
```

```
## model to test simpleswap reps 100
```

```
##
```

```
## Model=      simpleswap
```

```
## Branchs swapped=  all
```

```
##
```

```
## BestInitial:A1
```

```
## AreaSelected Freq Percent
```

```
## 1          A1  73      73
```

```
## 2          A1A2  9       9
```

```
## 3          A2  18      18
```

```
##
```

```
##
```

```
##
```

```
## model to test allswap reps 100
```

```
##
```

```
## Model=      allswap
```

```
## Branchs swapped=  terminals
```

```
##
```

```
## BestInitial:A1
```

```
## AreaSelected Freq Percent
```

```
## 1          A1  51      51
```

```
## 2          A2  49      49
```

```
##
```

```
##
```

```
##
```

```
## model to test allswap reps 100
```

```

##
##   Model=      allswap
##   Branchs swapped=  internals
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1  100      100
##
##
##
## model to test allswap reps 100
##
##   Model=      allswap
##   Branchs swapped=  all
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   33      33
## 2          A1A2   37      37
## 3           A2   30      30
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  terminals
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   55      55
## 2           A2   45      45
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  internals
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1  100      100
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  all
##
## BestInitial:A1
##   AreaSelected Freq Percent
## 1           A1   55      55

```

```
## 2          A2    45    45
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")
```

```
printEvalBranch(AllTerminals, compact=FALSE)
```

```
##      node initialArea FinalArea Approach %Delta
## 1 [t1]|1          A1          A2    lower -90.096
## 2 [t2]|2          A1          A1    lower      0
## 3 [t3]|3          A1          A1    lower      0
## 4 [t4]|4          A1          A1    lower      0
## 5 [t1]|1          A1          A1    upper      0
## 6 [t2]|2          A1          A2    upper 910.004
## 7 [t3]|3          A1          A1    upper      0
## 8 [t4]|4          A1          A2    upper 910.004
```

Now the internals

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")
```

```
printEvalBranch(AllTerminals, compact=FALSE)
```

```
##      node initialArea FinalArea Approach %Delta
## 1 [node number:6:t1/t2]|6          A1          A1    lower      0
## 2 [node number:7:t3/t4]|7          A1          A1    lower      0
## 3 [node number:6:t1/t2]|6          A1          A1    upper      0
## 4 [node number:7:t3/t4]|7          A1          A1    upper      0
```

Now all branches

```
Allall <- evalBranch(tree=initialTree,
                    distribution = dist4taxa,
                    branchToEval="all",
                    approach="all")
```

```
printEvalBranch(Allall, compact=FALSE)
```

```
##      node initialArea FinalArea Approach %Delta
## 1 [node number:6:t1/t2]|6          A1          A1    lower      0
## 2          [t1]|1          A1          A2    lower -90.096
## 3          [t2]|2          A1          A1    lower      0
## 4 [node number:7:t3/t4]|7          A1          A1    lower      0
## 5          [t3]|3          A1          A1    lower      0
## 6          [t4]|4          A1          A1    lower      0
## 7 [node number:6:t1/t2]|6          A1          A1    upper      0
## 8          [t1]|1          A1          A1    upper      0
## 9          [t2]|2          A1          A2    upper 910.004
## 10 [node number:7:t3/t4]|7          A1          A1    upper      0
```



```
## 11          [t3] |3          A1          A1  upper          0
## 12          [t4] |4          A1          A2  upper 910.004
```

#now with a longer branch

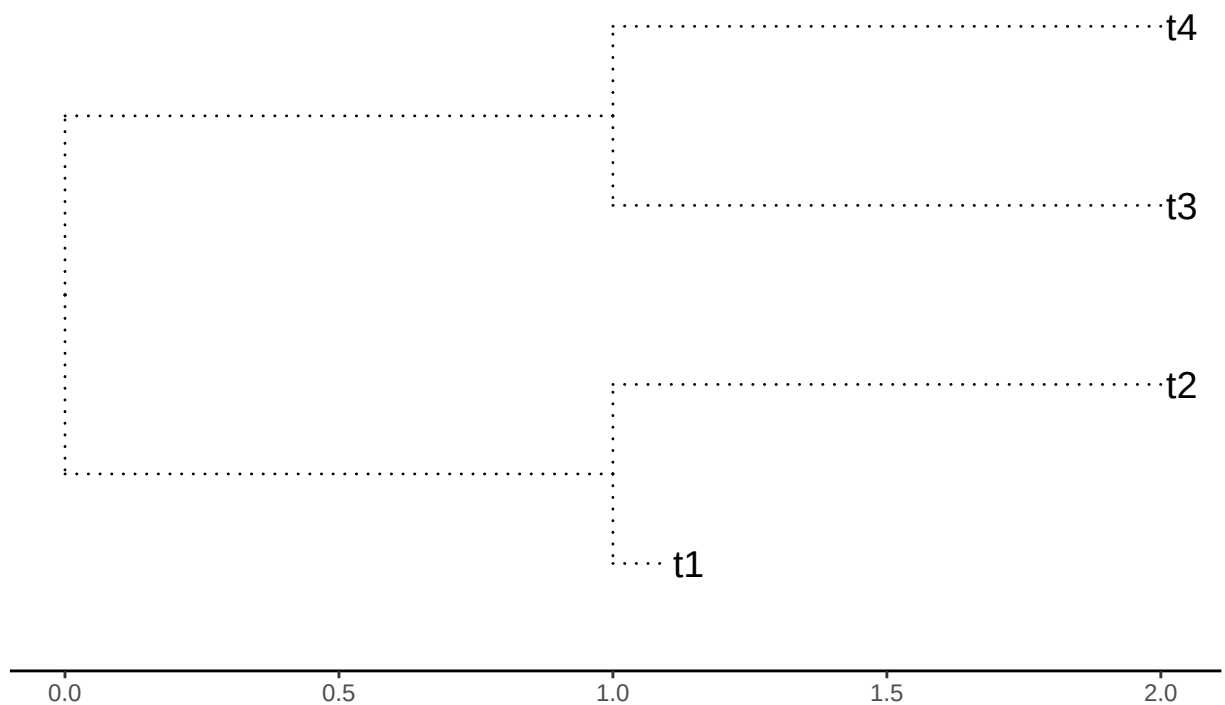
```
initialTree <- tree

initialTree$edge.length[2] <- 0.1

plotTree <- ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
  geom_tiplab(size=5, color="black") +
  theme_tree2() +
  ggtitle("Four terminals branch 1 == 0.1")

##
print(plotTree)
```

Four terminals branch 1 == 0.1



```
for( modelo in c("simpleswap", "allswap", "uniform") ){
  for( rama in c("terminals", "internals", "all") ){
    val <- swapBL(tree=initialTree,
```

```

        distribution = dist4taxa,
        model = modelo,
        branch = rama
    )

    cat( "\n\t Model=\t\t",modelo,
        "\n\t Branchs swapped=\t",rama,"\n" )

    printBlepd(val)

    cat( "\n\n\n" )

}

}

```

```

## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  terminals
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A1   27      27
## 2           A2   73      73
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  internals
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A2  100     100
##
##
##
## model to test simpleswap reps 100
##
##   Model=      simpleswap
##   Branchs swapped=  all
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A1   12      12
## 2          A1A2   18      18
## 3           A2   70      70
##
##
##
## model to test allswap reps 100
##

```

```

## Model=      allswap
## Branchs swapped=  terminals
##
## BestInitial:A2
## AreaSelected Freq Percent
## 1          A1  63      63
## 2          A2  37      37
##
##
##
## model to test allswap reps 100
##
## Model=      allswap
## Branchs swapped=  internals
##
## BestInitial:A2
## AreaSelected Freq Percent
## 1          A2 100     100
##
##
##
## model to test allswap reps 100
##
## Model=      allswap
## Branchs swapped=  all
##
## BestInitial:A2
## AreaSelected Freq Percent
## 1          A1  40      40
## 2          A1A2 33      33
## 3          A2  27      27
##
##
##
## model to test uniform reps 100
##
## Model=      uniform
## Branchs swapped=  terminals
##
## BestInitial:A2
## AreaSelected Freq Percent
## 1          A1  37      37
## 2          A2  63      63
##
##
##
## model to test uniform reps 100
##
## Model=      uniform
## Branchs swapped=  internals
##
## BestInitial:A2
## AreaSelected Freq Percent
## 1          A2 100     100

```

```
##
##
##
## model to test uniform reps 100
##
##   Model=      uniform
##   Branchs swapped=  all
##
## BestInitial:A2
##   AreaSelected Freq Percent
## 1           A1   56      56
## 2           A2   44      44

AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##      node initialArea FinalArea Aproach %Delta
## 1 [t1]|1           A2          A2  lower      0
## 2 [t2]|2           A2          A1  lower -90.115
## 3 [t3]|3           A2          A2  lower      0
## 4 [t4]|4           A2          A1  lower -90.115
## 5 [t1]|1           A2          A1  upper  910.412
## 6 [t2]|2           A2          A2  upper      0
## 7 [t3]|3           A2          A1  upper  91.905
## 8 [t4]|4           A2          A2  upper      0
```

Now the internals

```
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval="internals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##      node initialArea FinalArea Aproach %Delta
## 1 [node number:6:t1/t2]|6           A2          A2  lower      0
## 2 [node number:7:t3/t4]|7           A2          A2  lower      0
## 3 [node number:6:t1/t2]|6           A2          A2  upper      0
## 4 [node number:7:t3/t4]|7           A2          A2  upper      0
```

Now all branches

```
Allall <- evalBranch(tree=initialTree,
                     distribution = dist4taxa,
                     branchToEval="all",
                     approach="all")

printEvalBranch(Allall, compact=FALSE)
```

##	node	initialArea	FinalArea	Approach	%Delta
## 1	[node number:6:t1/t2] 6	A2	A2	lower	0
## 2	[t1] 1	A2	A2	lower	0
## 3	[t2] 2	A2	A1	lower	-90.115
## 4	[node number:7:t3/t4] 7	A2	A2	lower	0
## 5	[t3] 3	A2	A2	lower	0
## 6	[t4] 4	A2	A1	lower	-90.115
## 7	[node number:6:t1/t2] 6	A2	A2	upper	0
## 8	[t1] 1	A2	A1	upper	910.412
## 9	[t2] 2	A2	A2	upper	0
## 10	[node number:7:t3/t4] 7	A2	A2	upper	0
## 11	[t3] 3	A2	A1	upper	91.905
## 12	[t4] 4	A2	A2	upper	0

#now with a longer branch